

Fundamentals of Generative AI – Course Modules

Module 1: GenAI Foundations

1. Roadmap: GenAI Fundamentals
2. What is Generative AI?
3. Traditional AI/ML vs Generative AI
4. How LLMs Work (Without the Math)
5. Transformer Architecture – LLM Evolution
6. What Makes an LLM Powerful (Model Anatomy)

Module 2: LLM Mechanics & Ecosystem

1. LLM Ecosystem & Model Landscape
2. APIs vs Local Models (Real-World Decisions)
3. Tokens, Context Windows & Cost Mechanics
4. LLM Limitations & Failure Modes

Module 3: Building with LLMs

1. Prompting Fundamentals
2. Prompt Tuning vs Fine-Tuning
3. LoRA & QLoRA Intuition
4. Embeddings & Semantic Understanding
5. LLM Memory & Context Management
6. RAG Fundamentals
7. Tool Use & Function Calling
8. Agentic AI – Mental Models
9. When NOT to Use Agents

Module 4: Responsible & Practical GenAI

1. Responsible AI for GenAI Developers